

CR-016: Show / Hide Password Visibility Toggle in the Login Module

1. Change Request Details

Field	Details
Change Request ID	CR-016
Title	Show / Hide Password Visibility Toggle in the Login Module
Priority	Medium
Difficulty	Low
Change Type	Frontend Enhancement
Affected Version	Student Management System v1.1 baseline
Dependency	Existing Login Module (<code>LoginPage.jsx</code> , <code>LoginFormSection.jsx</code>)

2. Business Requirement

Users and administrators currently enter passwords into a masked input field without an option to view or verify entered text. As passwords often contain numbers and symbols, users frequently make typing errors without realizing it.

The lack of visibility increases failed login attempts and repeated entries. The system should provide an intuitive password toggle allowing users to view or mask credentials before submitting.

3. Current Behaviour

- `LoginPage.jsx` renders `LoginFormSection.jsx` with username and password.
- The password input is statically configured with `type="password"`.
- Password input characters are masked unconditionally with bullets.
- No visibility toggle button or eye icon exists to reveal entered text.
- Existing authentication sends credentials via `POST /api/auth/login`.

4. Requested Change

Add a client-side show/hide password toggle to the login form password input.

The implementation must:

- Toggle password input type between `"password"` and `"text"`.
- Default to masked/hidden state upon component mount and page load.
- Display intuitive eye and eye-slash visual indicator icons.
- Include accessible `aria-label` and `aria-pressed` attributes.
- Ensure toggle button uses `type="button"` to prevent form submission.
- Adjust layout in `login.css` so password text does not overlap the icon.
- Support keyboard activation via Space and Enter keys.
- Preserve existing login validation and error alert banners.
- Pass entered credentials accurately to `login(username, password)`.
- Maintain existing administrator and student authentication flows.

5. Scope

In Scope

- Login password visibility toggle button and icon.
- Client-side visibility state management in `LoginFormSection.jsx`.
- Dynamic input type switching between "password" and "text".
- Keyboard accessibility and screen reader labeling.
- Password input wrapper and toggle button styling.
- Adequate input right padding to preserve text readability.
- Responsive layout across desktop, tablet, and mobile screens.
- Preservation of existing login submission and error handling.
- Retention of existing authentication and navigation flows.

Out of Scope

- Database changes or user table schema modifications.
- Backend API or controller changes.
- New API endpoints or authentication token changes.
- Server-side password hashing or verification changes.
- Password complexity or strength indicators.
- Password reset or Forgot Password workflows.
- Changes to student dashboard, admin dashboard, or student CRUD logic.

6. Expected / Potentially Affected Files

- `frontend/src/components/LoginFormSection.jsx`
- `frontend/src/styles/login.css`
- `frontend/src/pages/LoginPage.jsx` (optional)

Students may modify additional files if required. Any additional files must be identified and justified in the Implementation Report.

7. Functional & Security Requirements

- **FR-001:** Visibility toggle button shall be provided on the password input.
- **FR-002:** Activating toggle shall switch between masked and visible text.
- **FR-003:** Password field shall default to masked state when page renders.
- **FR-004:** Visual icon shall clearly reflect current state (eye / eye-slash).
- **FR-005:** Toggle button shall have `type="button"` to prevent form submit.
- **FR-006:** Toggle button shall be operable via keyboard (Tab, Space, Enter).
- **FR-007:** Existing login submission, validation, and alerts shall function.
- **FR-008:** Role-based navigation to Admin or Student Dashboard shall persist.
- **SEC-001:** Plaintext passwords shall not be persisted in browser storage.
- **SEC-002:** Default masked state shall protect against shoulder-surfing.

8. Acceptance Criteria

The CR is successfully implemented when:

- **AC-001:** Users can toggle password visibility between hidden and visible.
- **AC-002:** Default state of the password field on load is masked.
- **AC-003:** Toggle icon updates dynamically to match visibility state.
- **AC-004:** Activating the toggle does not trigger login form submission.
- **AC-005:** Toggle button is keyboard accessible and includes aria-label.
- **AC-006:** Password text remains legible and is not obscured by the toggle icon.
- **AC-007:** Valid credentials submitted in either state log in successfully.
- **AC-008:** Existing validation errors and role-based redirects continue to work.

9. Expected Outcome

Users can verify entered credentials before submitting the login form, reducing authentication errors and improving login usability, while existing backend APIs, authentication services, security mechanisms, and dashboard routing remain completely intact.

10. Implementation Report Requirement

After implementation, the Final Exam Software Change Implementation Report shall include:

- Implementation summary.
- Affected files.
- Important technical/security changes.
- Test cases and results.
- Acceptance criteria verification.
- Relevant screenshots/evidence.
- Final status and outstanding issues.